

Autoimmunity



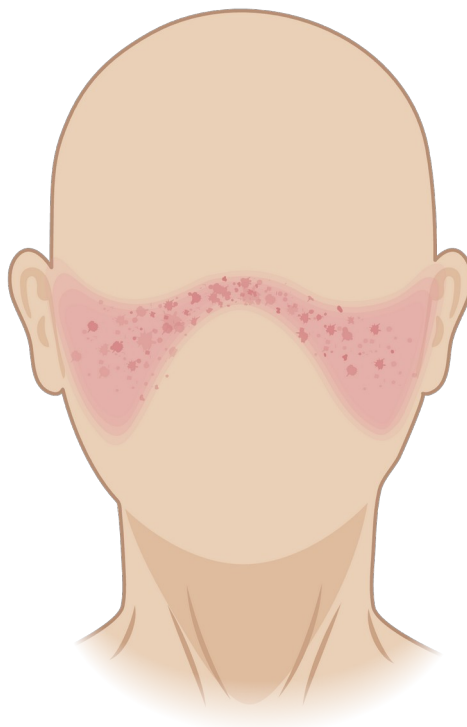
Clinical Features



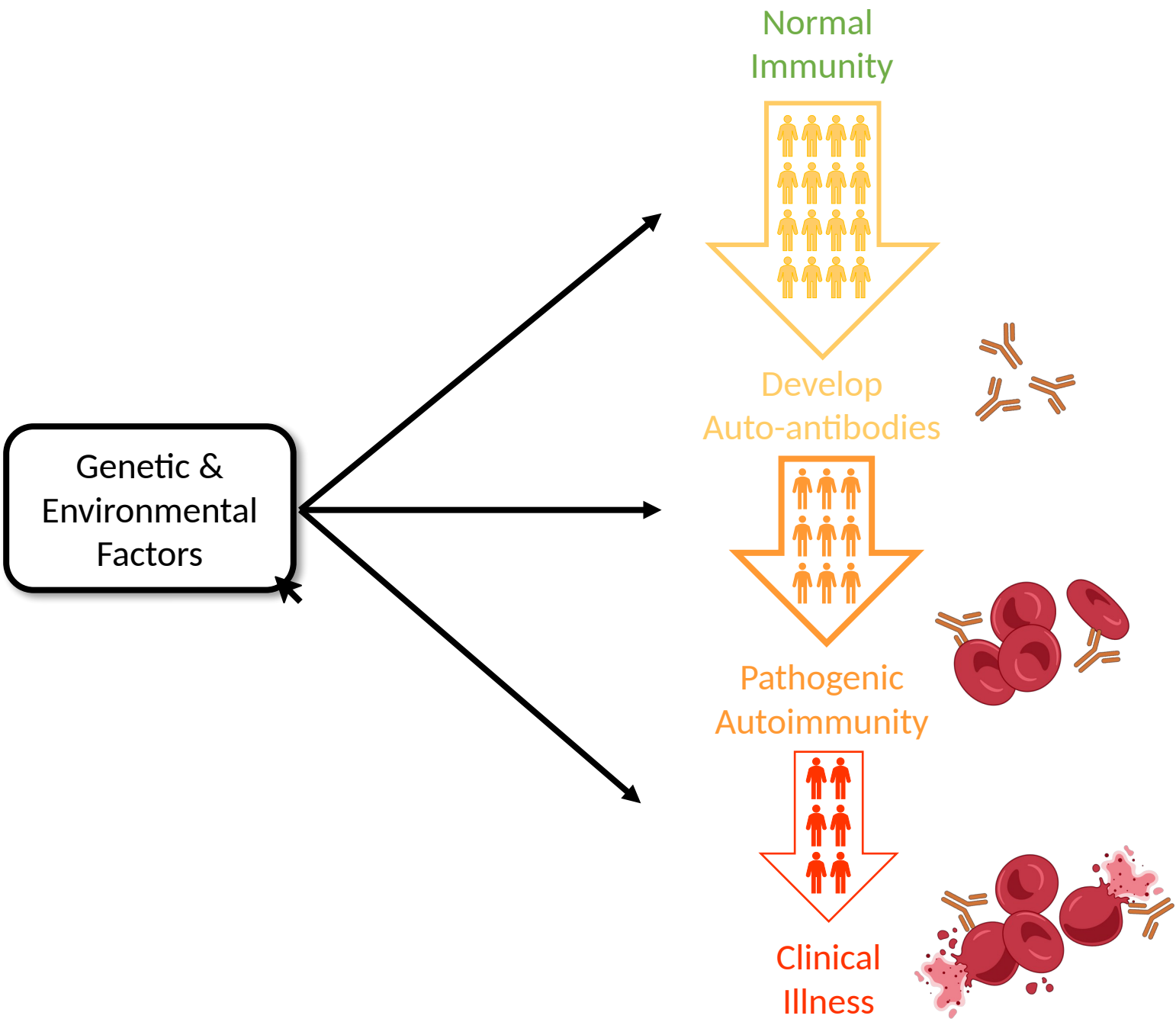
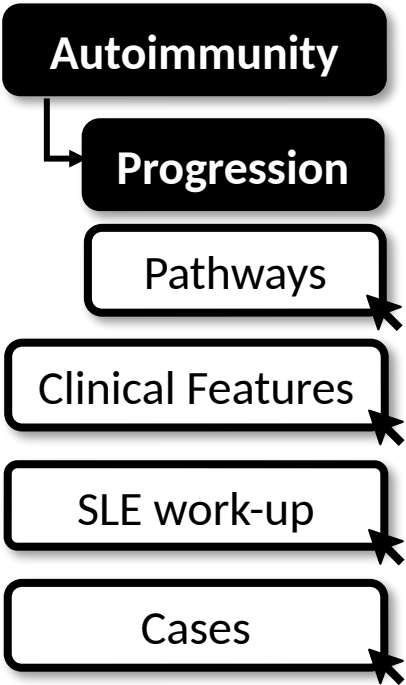
SLE work-up



Cases



Is it lupus?



Autoimmunity

Progression

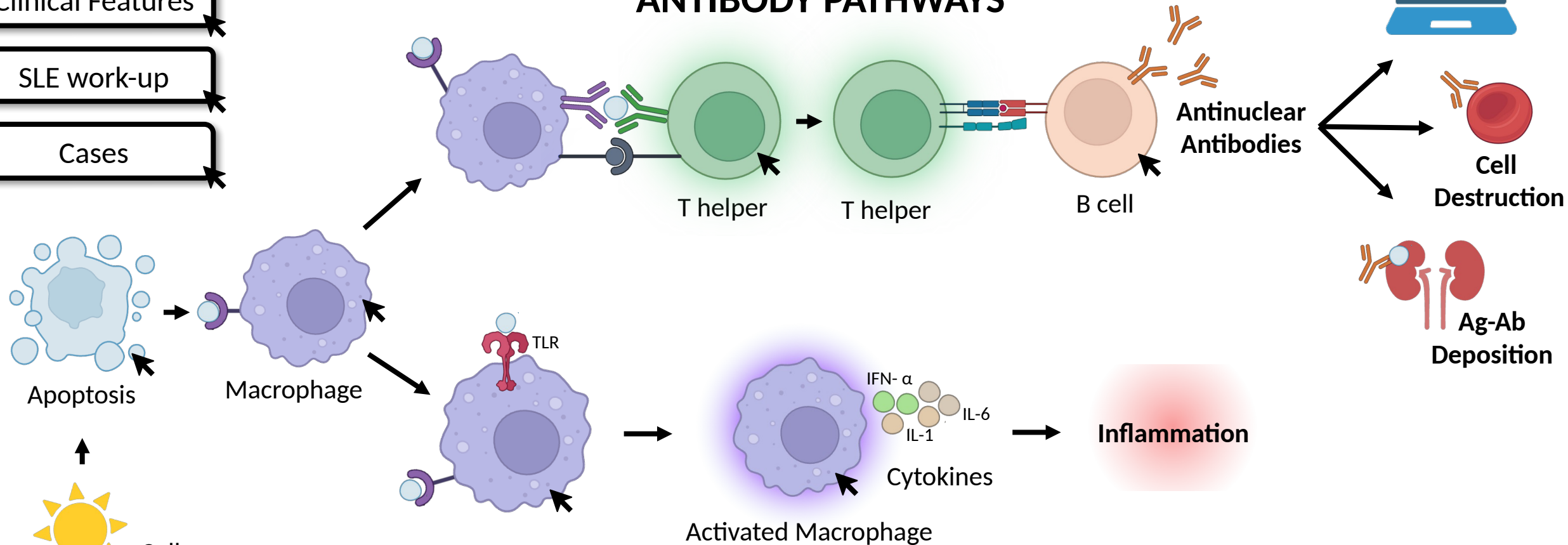
Pathways

Clinical Features

SLE work-up

Cases

ANTIBODY PATHWAYS



INFLAMMATORY PATHWAYS

Autoimmunity

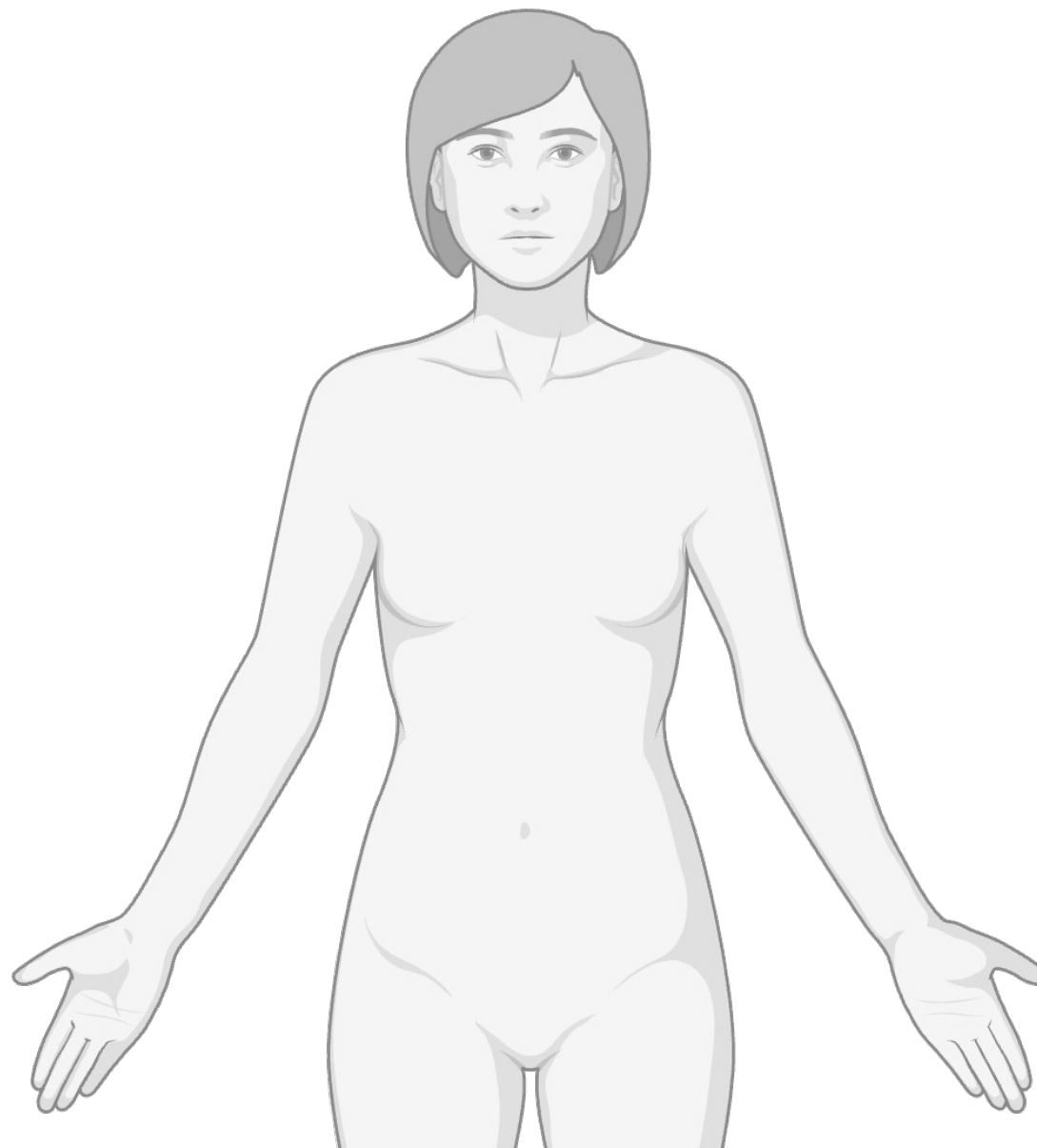
Clinical Features

SLE work-up

Cases

Concerning for lupus

LESS concerning for lupus



Autoimmunity

Clinical Features

SLE work-up

Cases

Concerning for lupus

LESS concerning for lupus

Warrants SLE testing, if not otherwise explained

Constitutional

Unexplained fevers

Mucocutaneous

Alopecia
Oral + nasal ulcers
Sun-sensitive rashes
Raynaud's
Sicca

Pulmonary

Interstitial lung disease
Pulmonary HTN

Renal

Proteinuria
Hematuria

Hematologic

Cytopenias
VTE

Neuropsychiatric

Psychosis
Seizures

Serosal

Pleural effusion
Pericardial effusion
Acute pericarditis

Gastrointestinal

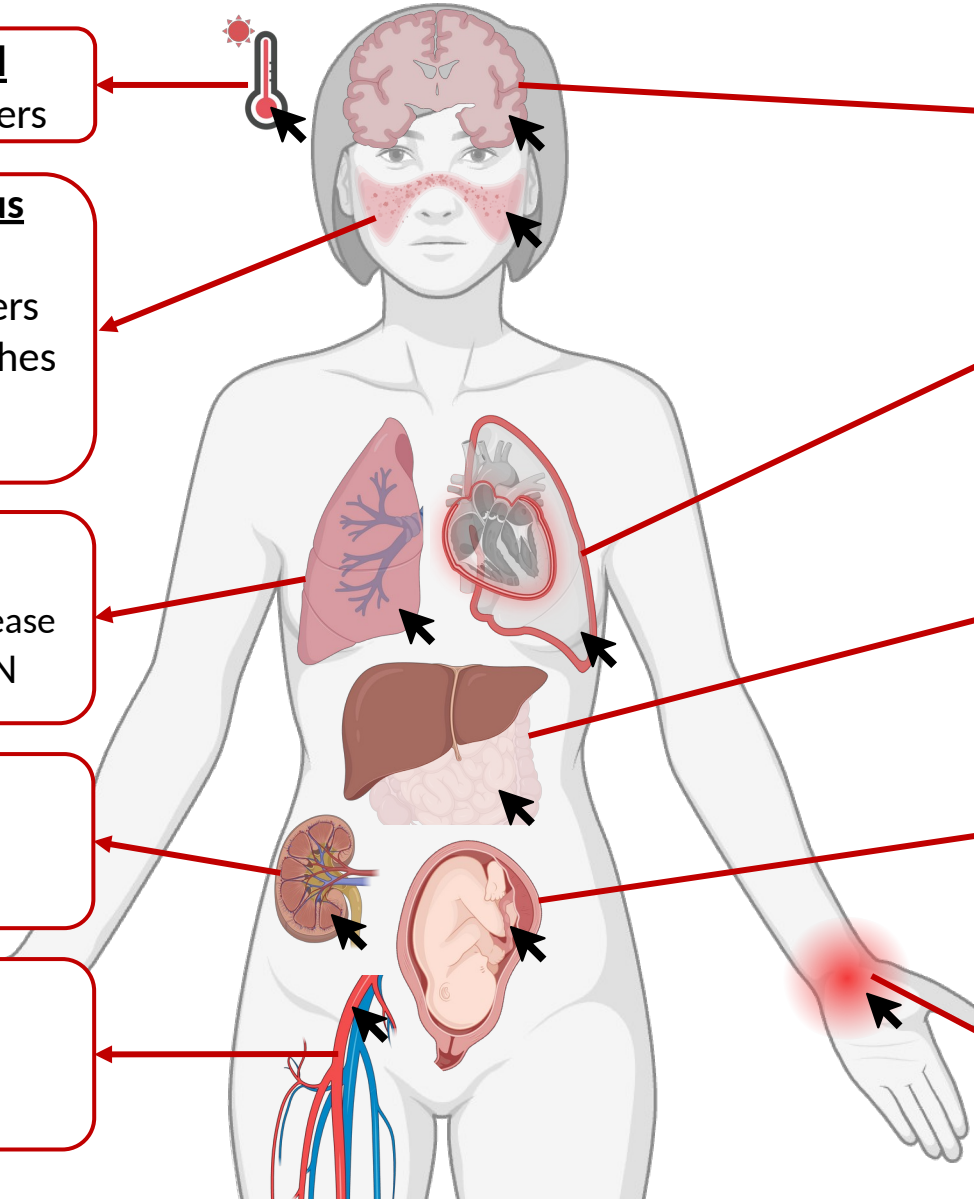
Autoimmune hepatitis
Mesenteric Vasculitis

Genitourinary

Recurrent and late
miscarriage

Musculoskeletal

Synovitis
Myositis



Autoimmunity

Clinical Features

SLE work-up

Cases

Concerning for lupus

LESS concerning for lupus

Should NOT prompt an SLE work-up

Constitutional

Fatigue
Weight changes

Mucocutaneous

Hair loss in shower

Pulmonary

Dyspnea

Renal

Pyuria

Hematologic

Iron deficiency
anemia

Neuropsychiatric

Anxiety, Depression
Headaches
Brain fog

Serosal

Atypical chest pain

Gastrointestinal

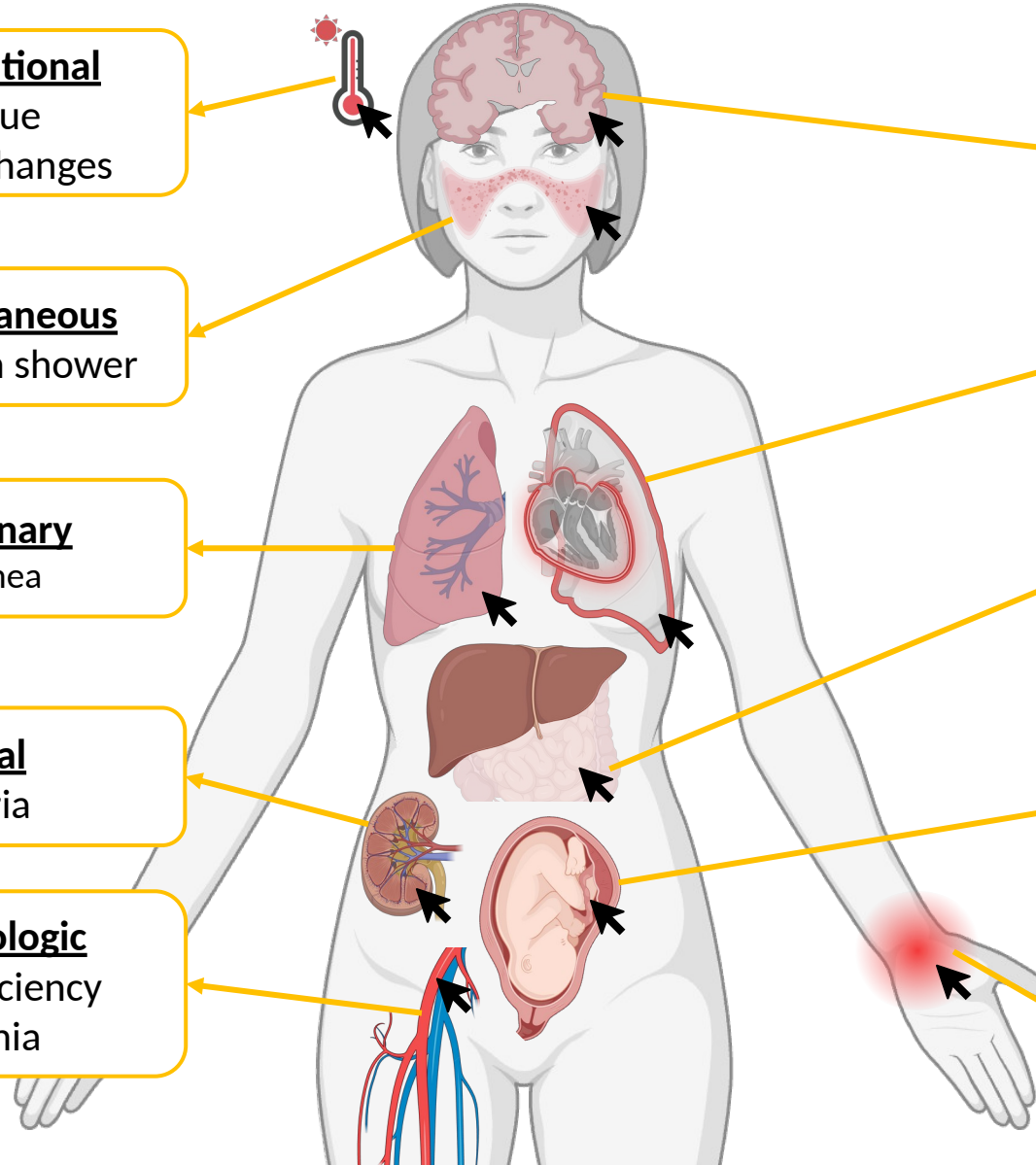
Abdominal pain
Chronic diarrhea
Hematochezia

Genitourinary

<2 miscarriages or
before 10 wks

Musculoskeletal

Generalized aches
Pain w/o swelling



What testing should be done to evaluate for suspected SLE?

Autoimmunity

Clinical Features

SLE work-up

Get an ANA?

Cases

CBC w/ differential

- Anemia
- Leukopenia
- Lymphopenia
- Neutropenia
- Thrombocytopenia

CMP

Elevated Cr
Transaminitis
Hyperbilirubinemia

Complement proteins

↓ C3 and/or C4

Inflammatory Markers

ESR >> CRP

Autoantibodies

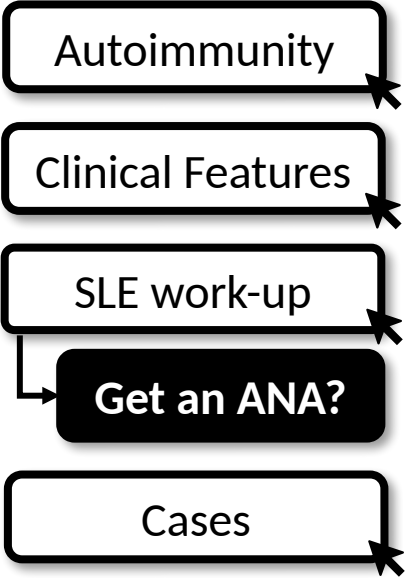
ANA
Anti-dsDNA Ab
Anti-Smith Ab
Anti-RNP Ab
Anti-SSA Ab
Anti-SSB Ab

Antiphospholipid antibodies

Anti-cardiolipin
Anti- β 2 glycoprotein
Lupus anticoagulant

UA

Hematuria
Proteinuria



Should an ANA be ordered?

37-year-old woman presents to the ED with chest pain. She has been recovering from COVID-19. Her VS are normal aside from HR 110. Her troponins peaked at 0.04 and 0.03. ECG shows ST segment elevation in all leads. She is diagnosed with pericarditis and started on colchicine and NSAIDs.

ANA Case 1

?

The patient's chest pain resolves, but one year later she returns with similar symptoms and is diagnosed with recurrent pericarditis. Her labs are notable for a normal WBC but with a low ANC.

ANA Case 1b

?

32-year-old woman with a h/o GAD, MDD and IBS presents to her PCP with a constellation of symptoms over the past year, including: fatigue, diffuse bilateral leg pain, neck pain, headaches, weight loss and diffuse body stiffness.

ANA Case 2

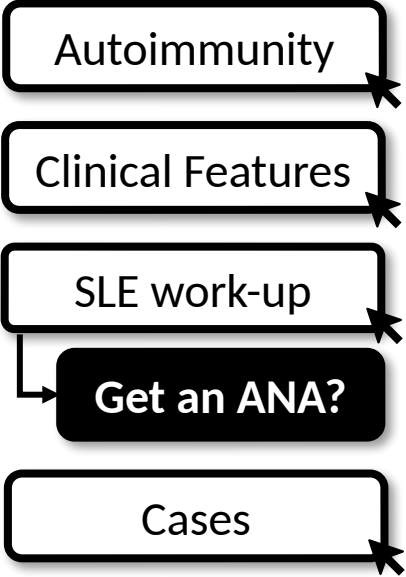
?

50-year-old woman with alcohol use disorder presents to the ED with fatigue and severe abdominal pain. Labs show pancytopenia: WBC 2.9, Hgb 9.7, Plt 56. Her AST/ALT are 12 and 15. T-Bili is 4.1. EtOH level is 124 (H) and UTox is negative. One day after admission, she develops visual hallucinations.

ANA Case 3

?





Should an ANA be ordered?

27-year-old male soccer player presents to his PCP for joint pain. He has noted pain and weakness in his left knee for 1 week, making it difficult for him to train. On exam, his VS are normal, his joints are not tender to palpation, and you note a raised rash on his face and arms. His CK is elevated at 12,000. An MRI of thighs demonstrates muscle edema.

ANA Case 4

?

19-year-old female college student presents to her PCP for fatigue. She is on a pre-med track and has become unable to attend class due to extreme exhaustion. She was told she has mono by the student health clinic after she was found to have a T 101. She has been laying in bed for weeks, her arms and legs are starting to feel sore. On exam, she is afebrile, has normal-appearing skin over her extremities, has a normal gait and normal ROM of her knees and elbows, but is noted to have tender and swollen wrists.

ANA Case 5

?

32-year-old man with a history of stimulant use disorder is brought to the ED by EMS after being found unresponsive. On exam, he has an erythematous, swollen left eye. Labs are notable for a Cr 2.3 and WBC 1.7. On a trauma scan, he is found to have splenic infarct.

ANA Case 6

?

Autoimmunity

Clinical Features

SLE work-up

Cases

1

2

3

Case 1

27-year-old woman with bilateral ankle pain and swelling. She is physically active, though has had fatigue limiting her ability to run outside. She also notes a rash on her face and extremities that is worse in the sun, as well as a rash on her arm that has not gone away. She has no other joint pain and denies fevers, chest pain, hair loss or oral ulcers. She has been told in the past that her white blood cell count is low. On exam, she has ankle tenderness and warmth as well as a rash on her arms.



Which of her signs or symptoms are most concerning for lupus ?

- Symptoms (rash, fatigue, joint pain) flare in the sun
- Joint swelling, warmth on exam, not 'aches and pains'

What labs should be ordered to work up her symptoms?

137	114	13	105	3.0	13.2	221	Abs lymph 0.6 (L)
4.3	22	0.6					

ANA: Positive, 1:320 Fine Speckled
Anti-DS-DNA: 1:40 (H)
Antiphospholipid Abs: Negative
C3: 109 (nl), C4: 5 (L)
ESR: 6 (nl), CRP: 3 (nl)
UA: bland

What are your next steps in management?

1. Refer to Rheumatology for a **likely diagnosis of SLE**
2. Symptom journal, noting triggers
3. Request patient to take photos of any rashes
4. Counsel on daily sunscreen (SPF 30+) NSAIDs for joint pain
5. Encourage pregnancy prevention until plan established
6. Indications for urgent referral to Rheumatology
 - Evidence of end-organ damage (e.g. renal, neurologic, hepatic)
 - Red, hot, swollen joints on exam
 - Severe hematologic derangements
 - Extensive body surface area involvement of photosensitive rash
 - Rash complicated by bullae or signs of infection
 - If the patient is pregnant

Autoimmunity

Clinical Features

SLE work-up

Cases

1

2

3

Case 2

40-year-old woman with SLE c/b Class IV lupus nephritis who presents to the ED with fevers, fatigue, myalgias, chest pain and dyspnea for five days. Receives monthly cyclophosphamide infusions and has been on prednisone 40mg daily.

Vitals: T 102.5, P 110, BP 105/72, RR 24, O2 sat 94% on room air.

Exam: Normal heart sounds, decreased breath sounds L>R, bibasilar crackles on pulmonary exam, no skin rashes, no warmth/tenderness/erythema of any joints

Labs: WBC 3.2, Hgb 9.5, CRP 77, ESR 130.

Imaging: CXR with a LLL infiltrate and L>R pleural effusions.

What is the differential to explain this presentation?

Lupus disease activity:

- Lupus pneumonitis
- Pleural effusion
- Pulmonary vasculitis (diffuse alveolar hemorrhage)
- Pulmonary embolism

Non-autoimmune:

- Bacterial pneumonia
- Fungal pneumonia
- Pulmonary embolism
- Bacterial endocarditis

Treatment-related:

- Cyclophosphamide-induced lung injury

What are your next steps in management?

1. Standard empiric antimicrobial coverage
2. Continue prednisone at home dose
3. Consult pulmonary to discuss bronchoscopy

What key points:

1. Infections are a major cause of mortality in lupus patients
2. Risk factors include both lupus and immunosuppressants
3. Glucocorticoids decrease inflammatory cytokines (good for lupus) but also the activity of macrophages and neutrophils (bad for infection)
4. In a patient with SLE and high CRP, think infection first

Autoimmunity

Clinical Features

SLE work-up

Cases

1

2

3

Case 3

35-year-old woman with depression who presents to her PCP with polyarthralgias, skin rashes and fatigue for two years. Upon further history, her pain involves her wrists, shoulders, low back, knees and ankles as well as deep muscle pain in her arms and legs. When she goes for a walk, she feels like it requires all her energy. She also experiences skin rashes on her eyelids and neck that are itchy and worse in the winter. She struggles with fatigue daily regardless of how much she sleeps. Her PCP orders an autoimmune panel that is notable for a positive ANA of 1:80. The remainder of her labs are all normal.

What is the differential to explain this presentation?

- Depression
- Anxiety
- Fibromyalgia
- Chronic fatigue syndrome
- SLE
- SLE + fibromyalgia

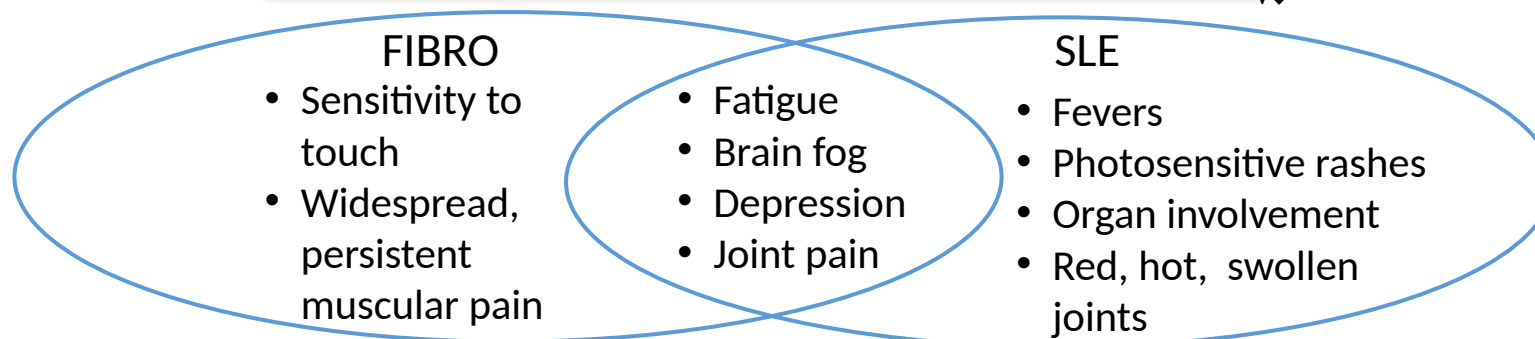
What percentage of individuals without SLE will have a positive ANA of 1:40? 1:80? 1:160?

1:40 = 32% | 1:80 = 13% | 1:160 = 5%

What percentage of individuals with SLE have fibromyalgia?

At least 20%

What features of fibromyalgia and SLE overlap?



Take-Home Points

1. SLE has a wide array of complications, but they trace back to a few immune pathways — autoantibody (ANA) production and TLR-driven inflammatory cytokines (IL-1, IL-6, IFN-alpha).
2. Before ordering an ANA, use a systems-based exam to look for SLE-SPECIFIC manifestations and judge the pre-test probability — an ANA in the wrong patient creates more problems than it solves.
3. "Yellow-box" symptoms (fatigue, diffuse aches/pain without swelling, anxiety/depression/brain fog, isolated atypical chest pain, hair loss in the shower, iron-deficiency anemia) should NOT by themselves trigger an ANA.
4. If suspicion is high enough to order an ANA, send the broader workup too: CBC w/ diff, CMP, C3/C4, ESR/CRP, dsDNA/Smith/RNP/SSA-SSB, antiphospholipid antibodies, and UA.
5. A low-positive ANA is common in healthy people (1:40 ~32%, 1:80 ~13%, 1:160 ~5%); fibromyalgia frequently coexists with SLE and explains many "yellow-box" symptoms.

2025 Context — Scope & Currency

This is a **DIAGNOSIS / ANA-stewardship** talk; its clinical-manifestation framework is built on the **EULAR/ACR 2019 SLE Classification Criteria**, which remain current (June 2026).

Diagnostic teaching unchanged. Beyond this talk's scope, note the TREATMENT landscape has shifted:

2024 ACR Lupus Nephritis guideline (Sammaritano, Arthritis Care Res 2025): a conceptual shift to early COMBINATION therapy — hydroxychloroquine for all; for active class III/IV($\pm$ V) LN, triple therapy = pulse + oral glucocorticoids PLUS (MPAA + belimumab) OR (MPAA + a calcineurin inhibitor, e.g. voclosporin) OR low-dose Euro-Lupus cyclophosphamide → MPAA + belimumab.

Lower-dose steroid taper: 0.5 mg/kg/day (max 40 mg) tapering to ≤ 5 mg/day by 6 months; add a RAAS inhibitor for proteinuria. KDIGO 2024 aligns.

Belimumab and voclosporin are both FDA-approved for lupus nephritis. Refer suspected/confirmed SLE to rheumatology (and nephrology for LN).

Attribution & Changelog

Original: "Is it lupus?" (Diagnosing Systemic Lupus Erythematosus) — TeachIM.org, published May 2024.

Authors: Carly Conran, MD (UCSF) & JoAnn Zell, MD (Univ. of Colorado). Section ed.: Meredith Morcos, MD. Executive ed.: Brandon Fainstad, MD.

Source: https://teachim.org/teaching_material/diagnosing-lupus/ | License: CC BY-NC 4.0 (non-commercial, attribution required).

Optimized 2026-06-11 (deck: Is-it-lupus-v2.pptx; all 12 original interactive slides preserved).

Correction: Case 1 labs (slide 10) "Antiphospholipid Abs" -> "Antiphospholipid Abs" (slide 7 already spelled correctly). Single XML run; all click-to-reveal animations preserved.

Currency check (June 2026): clinical framework verified vs EULAR/ACR 2019 Classification Criteria — unchanged. Treatment context (2024 ACR Lupus Nephritis guideline) added as out-of-scope note only.

Companion: SLE-Learner-Handout.pdf copied into the lesson folder and linked (reviewed, unchanged).